ASHLEY (YE) GAO

Curriculum Vitae

Assistant Professor, Department of Computer Science, William & Mary

540 Landrum Drive, Williamsburg, VA 23185 • ygao18@wm.edu • https://lindagaw.github.io

Academic Appointments

Assistant Professor, Department of Computer Science, William & Mary 2023–present

Education

Ph.D., Computer Science, University of Virginia 2019–2023

Advisor: John A. Stankovic. Dissertation: "Addressing Realisms Faced by Deep Learning Models in Cyber Physical Systems."

M.S., Computer Science, University of Virginia 2017–2019

B.S., Computer Science, University of California, San Diego 2012–2017

B.A., Literatures of the World, University of California, San Diego 2012–2017

Research Interests

I develop robust machine learning methods for affective computing and audio-language modeling. My work focuses on speech emotion recognition, audio large language models, and controllable text-to-speech.

- Domain-generalized audio sensing — robust speech emotion recognition under distribution shift
- Sentiment steering in large language models — controllable generation of emotional tone
- Audio stress detection for ADRD caregivers — mobile and scalable systems for caregiver support

Research Publications

† denotes graduate students advised by me: Jiacheng Shi, Hongfei Du, and Collin Macdonald (at the time of publication).

Refereed Conference and Workshop Proceedings

C18. Du, H.†, Shi, J.†, Zhang, Y., & **Gao, Y.** (2026). Tracing a sparse emotion-control circuit in LLM-based text-to-speech. Proceedings of EMNLP 2026.

C17. Du, H.†, Shi, J.†, Zhang, Y., Zhou, G., & **Gao, Y.** (2026). Usage-modulated sentiment representations in large language models. Proceedings of EMNLP 2026.

C16. Du, H.†, Shi, J.†, Lu, S., Zhou, G., & **Gao, Y.** (2026). Sparse autoencoders for interpretable emotion control in text-to-speech. Proceedings of ICML 2026, Proceedings of Machine Learning Research.

C15. Shi, J.†, Du, H.†, Song, X., Hong, Y. A., Zhang, Y., & **Gao, Y.** (2026). Emo-BPO: Emotion bidirectional preference optimization for diffusion-based emotional TTS. Proceedings of Interspeech 2026. (Forthcoming.)

C14. Shi, J.†, Du, H.†, Song, X., Hong, Y. A., Zhang, Y., & **Gao, Y.** (2026). AffectCodec: Emotion-preserving neural speech codec for expressive speech modeling. Findings of ACL 2026, pp. 9102–9124.

C13. Nair, T., Nafee, M. W., Jiang, M., **Gao, Y.**, Chen, H., & Zhang, Y. (2026). Confidence-aware ranker ensembles for robust in-context knowledge editing. Findings of ACL 2026, pp. 35070–35080.

C12. Shi, J.†, Du, H.†, He, Y., Hong, Y. A., & **Gao, Y.** (2026). Emotion-aligned generation in diffusion text-to-speech models via preference-guided optimization. Proceedings of ICASSP 2026.

C11. Shi, J.†, Du, H.†, Hong, Y. A., & **Gao, Y.** (2026). EMO-TTA: Improving test-time adaptation of audio-language models for speech emotion recognition. Proceedings of ICASSP 2026.

C10. Shi, J.†, Du, H.†, Hong, Y. A., & **Gao, Y.** (2026). Plug-and-play emotion graphs for compositional prompting in zero-shot speech emotion recognition. Proceedings of ICASSP 2026.

C9. Mersa, O., Jiang, M., **Gao, Y.**, Li, Q., & Zhang, Y. (2026). Hierarchical convolution multibranch transformer for EEG signals. Proceedings of ICASSP 2026.

C8. Jiang, M., Chen, Y., Liu, G., Sahu, A., **Gao, Y.**, & Zhang, Y. (2026). Pairs, not labels: Predicting protein-phenotype associations via link prediction. Proceedings of ICASSP 2026.

C7. Du, H.†, Shi, J.†, Myerston, J., Lu, S., Zhou, G., & **Gao, Y.** (2025). Role-guided annotation and prototype-aligned representation learning for historical literature sentiment classification. Findings of EMNLP 2025, pp. 3756–3768.

C6. Du, H.†, Lu, S., Zhou, G., & **Gao, Y.** (2025). EAA: Emotion-aware audio large language models with dual cross-attention and context-aware instruction tuning. Proceedings of Interspeech 2025, pp. 5433–5437.

C5. Shi, J.†, Zhang, Y., & **Gao, Y.** (2025). CLEP-DG: Contrastive learning for speech emotion domain generalization via soft prompt tuning. Proceedings of Interspeech 2025, pp. 4498–4502.

C4. Clapham, J., Zhou, M., MacDonald, C., Koltermann, K., **Gao, Y.**, & Shao, H. (2025). ElectroMeter: The practical electrolyte measurement system. Proceedings of CHASE 2025.

C3. Koltermann, K., Clapham, J., Blackwell, G., Jung, W., Burnet, E. N., **Gao, Y.**, Shao, H., Cloud, L., Pretzer-Aboff, I., & Zhou, G. (2024). Gait-Guard: Turn-aware freezing of gait detection for non-intrusive intervention systems. Proceedings of CHASE 2024, pp. 61–72. **Best Paper Award.**

C2. **Gao, Y.**, Baucom, B., Gordon, K. C., Rose, K. M., Wang, H., & Stankovic, J. A. (2023). E-ADDA: Unsupervised adversarial domain adaptation enhanced by a new Mahalanobis distance loss for smart computing. Proceedings of SMARTCOMP 2023, pp. 172–179. **Best Paper Runner-Up.**

C1. **Gao, Y.**, Ma, M., Gordon, K. C., Rose, K. M., Wang, H., & Stankovic, J. A. (2020). A monitoring, modeling, and interactive recommendation system for in-home caregivers: Demo abstract. Proceedings of SenSys 2020, pp. 587–588.

Refereed Journal Articles

J6. Ko, E., Rose, K. M., Gordon, K. C., Kim, S., **Gao, Y.**, Wang, P., Wijayasingha, L., Wang, H., Stankovic, J. A., & Wright, K. D. (2026). Feasibility and acceptability of the smarthealth intervention for dementia caregivers: A qualitative analysis of a single-group pilot study. Journal of Advanced Nursing, 82, 3891–3905.

J5. Ko, E., **Gao, Y.**, Wang, P., Wijayasingha, L., Wright, K. D., Gordon, K. C., Wang, H., Stankovic, J. A., & Rose, K. M. (2025). Recruitment challenges and strategies in a technology-based intervention for dementia caregivers: Descriptive study. JMIR Formative Research, 9(1), e59291.

J4. Macdonald, C.†, Chu, Z., Stankovic, J. A., Zhou, G., Shao, H., & **Gao, Y.** (2024). MiddleGAN: Generating inter-domain images with GANs. Transactions on Machine Learning Research (TMLR), 2024.

J3. **Gao, Y.**, Salekin, A., Gordon, K. C., Rose, K. M., Wang, H., & Stankovic, J. A. (2021). Emotion recognition robust to indoor environmental distortions and non-targeted emotions using out-of-distribution detection. ACM Transactions on Computing for Healthcare (HEALTH), 3(2), 1–22.

J2. **Gao, Y.**, Jabbour, J., Schlegel, E. C., Ma, M., McCall, M., Wijayasingha, L., Ko, E., Rose, K. M., Wang, H., & Stankovic, J. A. (2021). Out-of-the-box deployment to support research on in-home care of Alzheimer's patients. IEEE Pervasive Computing, 21(1), 37–47.

J1. Rose, K. M., Coop Gordon, K., Schlegel, E. C., McCall, M., **Gao, Y.**, Ma, M., Wijayasingha, L., Stankovic, J. A., & Wang, H. (2021). Smarthealth technology study protocol to improve relationships between older adults with dementia and family caregivers. Journal of Advanced Nursing, 77(5), 2519–2529.

Research Funding

Agentic AI for Multi-Modal Perception Integrity: Threat Reasoning and Response in Cyber-Physical Transportation Systems.

Co-Principal Investigator | Commonwealth Cyber Initiative, Coastal Virginia (COVA CCI) | 2026

Total award: $80,000; W&M allocation: $39,199. Principal Investigator: Qing Tang, Old Dominion University.

Teaching and Mentoring

All courses listed below were taught at William & Mary.

Graduate Courses

CSCI 680: Affective Computing
Fall 2026

CSCI 680: Deep Transfer Learning
Spring 2024, Spring 2025, Fall 2025

CSCI 516: Fundamentals of Artificial Intelligence and Machine Learning
Fall 2023, Fall 2024, Spring 2026

Undergraduate Courses

CSCI 150 / COLL 150: Artificial Intelligence in Speculative Media and Literatures
Fall 2026

CSCI 416: Fundamentals of Artificial Intelligence and Machine Learning
Fall 2023, Fall 2024, Spring 2026

Current Doctoral Advisees in the Galatea Lab

Hongfei Du, Ph.D. student, William & Mary — 2024–present

First-author publications: ICML, EMNLP (Findings), Interspeech.

Jiacheng Shi, Ph.D. student, William & Mary — 2024–present

First-author publications: ICASSP, ACL (Findings), Interspeech.

Hyo Jin Jon, Ph.D. student, William & Mary — 2026–present

Siyuan Zhao, Ph.D. student, William & Mary — 2026–present

Current Undergraduate Advisees

Ryan Hong, University of Maryland — 2026–present

Meave Meyer, William & Mary — 2026–present

Former Undergraduate Advisees

Lilly Liu, William & Mary — initial placement: Carnegie Mellon University — 2024–2026

Service

Editorial Positions

Associate Editor, Elsevier Smart Health — 2024–present

Invited Grant Review Panels

Panelist, National Institutes of Health (NIH) — 2025

Panelist, National Institutes of Health (NIH) — 2024

Panelist, National Science Foundation (NSF) — 2024

Conference Organization

Demo & Poster Chair, IEEE/ACM Conference on Connected Health (CHASE) — 2024

Reviewing

IEEE Affective Computing	2026
ICML; AAAI; ICLR	2025
NeurIPS; AAAI; ICLR; ACM Transactions on Computing for Healthcare	2024
AAAI; NeurIPS; ICLR	2023
IEEE Affective Computing; AAAI; ACM IMWUT/UbiComp; NeurIPS	2022
ACM Health	2021

Professional Memberships

Association for Computing Machinery (ACM)	2025
Institute of Electrical and Electronics Engineers (IEEE)	2025

Honors and Awards

Research and Service Awards

Best Paper Award, IEEE/ACM CHASE	2024
Best Service Award, IEEE/ACM CHASE	2024
Best Paper Runner-Up Award, IEEE SmartComp	2023
Best Poster Award, Computer Science Symposium, University of Virginia	2022

Fellowships and Academic Recognition

Ph.D. Fellowship, Department of Computer Science, University of Virginia	2019
Academic Excellence Fellowship, Department of Computer Science, University of Virginia	2017
Honors with High Distinction, University of California, San Diego	2016
Featured Honors Student, Department of Literature, University of California, San Diego	2016

Creative Writing

W2. **Gao, A.** (2026). The Tempest's Call. Pegasus Publishers, United Kingdom. (Forthcoming.)

W1. **Gao, A.** (2026). The purpose of a name. Neon Dystopia, United States. Short fiction.